INCH-POUND

MS75088D
W/AMENDMENT 1
7 October 2022
SUPERSEDING
MS75088D
31 August 2007

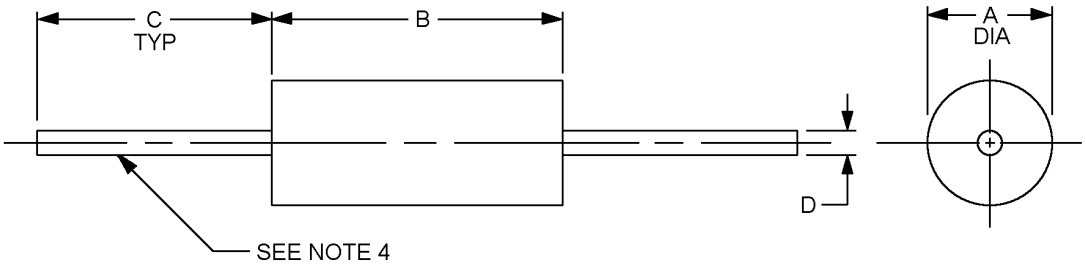
MILITARY SPECIFICATION SHEET

COILS, RADIO FREQUENCY, MOLDED, FIXED, MICRO-MINIATURE, MAGNETICALLY SHIELDED, (IRON CORE – IRON SLEEVE), TYPES LT10K203 TO LT10K216, INCLUSIVE

Inactive for new design, after 4 September 1985.
For new design, use MIL-PRF-39010/2.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the products described herein shall consist of this specification and MIL-PRF-15305.



LTR	Minimum	Maximum
A	.152 (3.86)	.172 (4.37)
B	.390 (9.91)	.430 (10.92)
C	1.250 (31.75)	1.626 (41.30)
D	.023 (0.58)	.027 (0.69)

NOTES:
1. Dimensions are in inches with metric equivalents (mm) in parentheses.
2. Metric equivalents are given for general information only.
3. These coils are intended to be supported by their leads.
4. Solderable/weldable lead wire, number 22 AWG.

FIGURE 1. Dimensions and configuration.

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REQUIREMENTS:

Design, construction, and physical dimensions: See figure 1.

Weight: 0.0353 ounce, maximum.

Operating temperature range: -55°C to +105°C.

Ambient temperature: + 90°C.

Temperature rise: 15 °C.

Terminal pull: 5 pounds, minimum

Altitude: 70,000 feet.

Shock (specified pulse): MIL-STD-202-213, test condition I, is applicable.

Dielectric withstanding voltage:

At sea level: MIL-STD-202-301, test voltage 1,000 V rms for a minimum of 60 seconds.

At reduced barometric pressure: MIL-STD-202-105, test condition C, test voltage of 200 V rms for a minimum of 60 seconds.

Percent coupling: 3 percent, maximum.

Electrical characteristics: See tables I and II.

Inductance: See table I.

Q values: See table I.

Self-resonant frequency (SRF): See table I.

DC resistance (DCR): See table I.

Part or Identifying Number (PIN): MS75088 - (dash number from table I).

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TABLE 1. Electrical characteristics (initial).

1/ Dash Number MS75088	Type Designation	Inductance (μH) ±10%	Q (min)	Test Frequency (MHz)	SRF Minimum (MHz)	DC resistance (ohms)	Rated DC current (mA)
-1	LT10K203	1.00	44	25	140	.07	1,070
-2	LT10K204	1.20	44	7.9	130	.10	895
-3	LT10K205	1.50	44	7.9	115	.12	815
-4	LT10K206	1.80	44	7.9	105	.14	775
-5	LT10K207	2.20	44	7.9	100	.19	650
-6	LT10K208	2.70	44	7.9	92	.28	535
-7	LT10K209	3.30	44	7.9	85	.35	480
-8	LT10K210	3.90	44	7.9	75	.40	450
-9	LT10K211	4.70	44	7.9	70	.55	380
-10	LT10K212	5.60	44	7.9	65	.72	335
-11	LT10K213	6.80	50	7.9	55	1.02	280
-12	LT10K214	8.20	50	7.9	50	1.32	250
-13	LT10K215	10.0	50	7.9	46	1.62	220
-14	LT10K216	12.0	55	2.5	44	2.00	200

1/ The coils specified herein are substitutes for the inactive coils on MS90537, providing the small decrease in physical dimensions are not a factor. The decrease in maximum operating temperature from 125°C to 105°C does not downgrade these coils but assures satisfactory operation at 105°C for a minimum of 2,000 hours of life, rather than a shorter period of operation at 125°C.

TABLE II. Electrical characteristics (final). 1/

Inspection group	Allowable variation from Initial measurement	Allowable variation from Initial measurement	Allowable percent from specified minimum value in electrical characteristics (initial) table	Allowable percent from specified minimum value in electrical characteristics (initial) table
	Inductance (percent)	DC resistance	Self-resonant frequency	Q
Qualification inspection				
Group II	±5	---	---	-10
Group III	±5	±(3% +.001 ohm)	-8	-10
Group IV	±5	±(2% +.001 ohm)	-5	-10
Conformance inspection group C				
Subgroup I	±5	---	---	-10
Subgroup II	±5	±(2% +.001 ohm)	-5	-10
Subgroup III	±5	±(3% +.001 ohm)	-8	-10

1/ Test fixture allowance of +.01μH shall be added to all change in inductance limits ±(_ percent +.01 μH).

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Referenced documents. In addition to MIL-PRF-15305, this document references the following:

MIL-STD-202-105 MIL-STD-202-213 MIL-STD-202-301

Custodians:
Army – CR
Navy - EC
Air Force - 85
DLA – CC

Preparing activity:
DLA – CC

Project 5950-2022-067

Review activities:
Army – AR, CR4, MI
Navy – AS, MC, OS, SH
Air Force – 19

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